

WASP-36b

R-1104

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_170202 · created 2026-07-30T19:58:37+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457786.723 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.165 +/- 0.049
Transit depth (Rp/Rs)^2	2.73 +/- 1.61 [%]
Orbital Inclination (inc)	87.99 +/- 2.94
Airmass coefficient 1 (a1)	0.58 +/- 0.24
Airmass coefficient 2 (a2)	0.4 +/- 0.37
Scatter in the residuals of the lightcurve fit is	41.82 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	4.5
Transit Duration (day)	0.0694 +/- 0.0052

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.165 ± 0.049 vs 0.1368 (deviation 0.58σ)
Duration fit vs published	0.0694 d vs 0.0758 d (deviation 1.24σ)
Mid-time O–C	5.6 min
Depth SNR	3.4
Residual scatter	41.82 %

Light curve

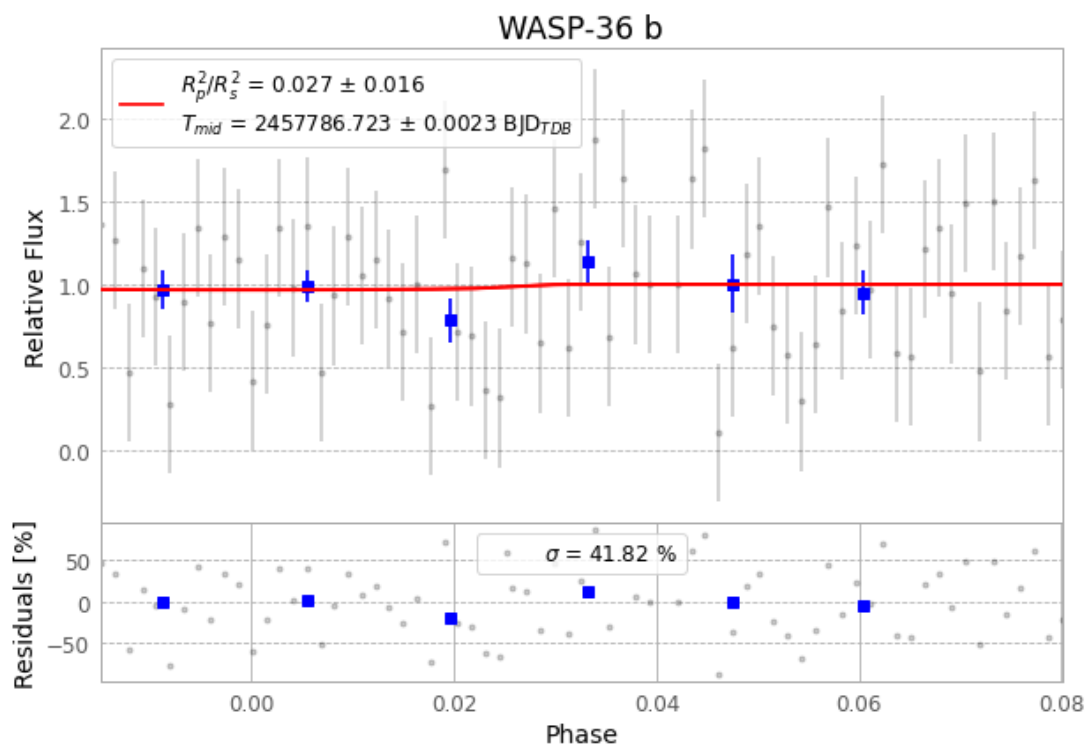


Figure 1 — FinalLightCurve_WASP-36 b_2017-02-01.png (SHA-256 6d927f77064ee080...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [198, 414], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 53aeecef6ae4e03d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

71 FITS frames (47 MB), WASP-36170202043912.FITS → WASP-36170202080912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-170202.log (08da20747898...), FinalLightCurve_WASP-36 b_2017-02-01.png (6d927f77064e...), FinalLightCurve_WASP-36 b_2017-02-01.csv (c4f214c6b5e6...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 19:58Z.